

Indefinite Kasparov modules and pseudo-Riemannian manifolds

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Abstract

We present a definition of indefinite Kasparov modules, a generalisation of unbounded Kasparov modules modelling non-symmetric and non-elliptic (e.g. hyperbolic) operators. Our main theorem shows that to each indefinite Kasparov module we can associate a pair of (genuine) Kasparov modules, and that this process is reversible. We present three examples of our framework: the Dirac operator on a pseudo-Riemannian spin manifold (i.e. a manifold with an indefinite metric); the harmonic oscillator; and the construction via the Kasparov product of an indefinite spectral triple from a family of spectral triples. This last construction corresponds to a foliation of a globally hyperbolic spacetime by spacelike hypersurfaces.

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1 Introduction

Both Connes' noncommutative geometry [Con94] and Kasparov's KK -theory [Kas80, BJ83] deal with non-commutative generalisations of elliptic, self-adjoint differential operators. As such, these frameworks are particularly suited to describe *Riemannian* manifolds. In this article we aim to extend these frameworks to allow for non-elliptic and non-symmetric operators, and in particular (normally) hyperbolic operators. Our motivating example is the Dirac operator on a *pseudo-Riemannian* manifold, i.e. a manifold equipped with an *indefinite* (but non-degenerate) metric. It is precisely this example that has inspired the terminology for the *indefinite* Kasparov modules we introduce in Definition 3.1.

Our definition of indefinite Kasparov modules is a generalisation of the usual definition of unbounded Kasparov modules [BJ83]. One of our main goals is to make sure that this generalised definition still allows us to remain in touch with all the usual tools of KK -theory [Kas80]. If $(\mathcal{A}, E_B, \mathcal{D})$ is an indefinite unbounded Kasparov module, we can construct from the (typically non-symmetric) operator $\mathcal{D}$ two symmetric operators given by

$$\mathcal{D}_{\pm} := \operatorname{Re} \mathcal{D} \pm \operatorname{Im} \mathcal{D} = \frac{1}{2}(\mathcal{D} + \mathcal{D}^*) \mp \frac{i}{2}(\mathcal{D} - \mathcal{D}^*).$$

We then want to make sure that these operators yield two unbounded Kasparov modules $(\mathcal{A}, E_B, \mathcal{D}_{\pm})$, and the main challenge here is to prove self-adjointness for $\mathcal{D}_{\pm}$.

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This article continues in the spirit of our previous paper [DPR13], where we defined *pseudo-Riemannian spectral triples* $(\mathcal{A}, \mathcal{H}, \mathcal{D})$ as a generalisation of spectral triples, and we showed that the operators $\mathcal{D}_\pm$ defined as above yield spectral triples. Although the motivation for the present article is the same, there are nonetheless several significant differences.

First, we work more generally with Kasparov modules instead of spectral triples. Second, while the definition of pseudo-Riemannian spectral triples requires assumptions on the *second-order* operators $\mathcal{D}\mathcal{D}^* + \mathcal{D}^*\mathcal{D}$ and $\mathcal{D}^2 - \mathcal{D}^{*2}$, the definition of indefinite Kasparov modules focuses more on *first-order* operators (namely $\mathcal{D}$, $\mathcal{D}^*$, $\operatorname{Re} \mathcal{D}$, and $\operatorname{Im} \mathcal{D}$), which is more natural. Third, the definition of indefinite Kasparov modules has the advantage that it does not require any smoothness properties. And fourth, it allows to *reverse* the procedure $\mathcal{D} \mapsto \mathcal{D}_\pm$, which means that we can characterise all pairs of unbounded Kasparov modules that can be obtained from an indefinite Kasparov module in this way.

As mentioned above, the main technical challenge is to obtain self-adjointness for $\mathcal{D}_\pm$. In [DPR13], this is achieved by assuming that $\langle \mathcal{D} \rangle^2 := (\operatorname{Re} \mathcal{D})^2 + (\operatorname{Im} \mathcal{D})^2$ is self-adjoint, and that the anti-commutator $\{\operatorname{Re} \mathcal{D}, \operatorname{Im} \mathcal{D}\}$ is ‘suitably bounded’ relative to $\langle \mathcal{D} \rangle^2$. In this article, we prefer to avoid assumptions on the second-order operator $\langle \mathcal{D} \rangle^2$. Instead, we now impose the condition that the real and imaginary parts of $\mathcal{D}$ *almost anti-commute*, which means that the anti-commutator $\{\operatorname{Re} \mathcal{D}, \operatorname{Im} \mathcal{D}\}$ is relatively bounded by $\operatorname{Re} \mathcal{D}$. A theorem of Kaad-Lesch [KL12] (quoted in Theorem 2.11) then allows us to conclude that $\mathcal{D}_\pm$ are self-adjoint.

Unfortunately, our main motivating example, namely the Dirac operator $\not{D}$ on a pseudo-Riemannian manifold, does not satisfy this condition. Indeed, although the anti-commutator $\{\operatorname{Re} \not{D}, \operatorname{Im} \not{D}\}$ is a first-order differential operator, it contains in general both spacelike derivatives and timelike derivatives, and thus it is not relatively bounded by $\operatorname{Re} \not{D}$ (nor by $\operatorname{Im} \not{D}$). In order to ensure that $\operatorname{Re} \not{D}$ and $\operatorname{Im} \not{D}$ almost anti-commute, we need the timelike part of $\{\operatorname{Re} \not{D}, \operatorname{Im} \not{D}\}$ to vanish identically, which places a restriction on the geometry of the pseudo-Riemannian manifold (see below). This asymmetry between the timelike and spacelike parts of $\{\operatorname{Re} \not{D}, \operatorname{Im} \not{D}\}$ is artificial, and indicates that it would be desirable to have a more general version of Kaad and Lesch’ theorem: we aim to return to this issue in a future work.

The layout of this article is as follows. In Section 2 we first describe our approach to dealing with non-symmetric operators, where we emphasise the real and imaginary parts of the operator. Subsequently, we gather some results on almost (anti-)commuting operators which will be useful later on.

Next, we define indefinite Kasparov modules as well as pairs of Kasparov modules in Section 3, and we prove our main theorem, which states that these definitions are equivalent. We continue in Section 3.2 by discussing the odd version of indefinite Kasparov modules. As for usual Kasparov modules, it is straightforward to turn an odd indefinite Kasparov module into an even one by ‘doubling it up’. We then prove that these odd modules are characterised by pairs of Kasparov modules for which the two operators are related via a certain unitary equivalence.

In Section 4 we discuss several examples. We start in Section 4.1 with the main motivating example, namely the Dirac operator on a pseudo-Riemannian spin manifold. We show, under certain mild assumptions on the manifold, that this Dirac operator satisfies all but one condition in the definition of indefinite Kasparov modules. The condition that fails (as mentioned above) is the assumption that the real and imaginary parts of the Dirac operator almost anti-commute in the sense of Kaad-Lesch. We continue to show that this condition does hold for the case of Lorentzian manifolds with ‘parallel time’. This example indicates that further study is required to obtain a more flexible formulation of indefinite Kasparov modules.

The second example (Section 4.2) considers the harmonic oscillator in arbitrary dimensions. This example in particular shows that manifolds with indefinite metrics are not the only examples of our framework.

Finally, in Section 4.3 we discuss families of spectral triples (building upon work by Kaad and Lesch [KL13]), and we show that one can naturally associate an indefinite Kasparov module to such families. Our work on families of spectral triples was initially motivated by the study of spacelike foliations of spacetime from the perspective of noncommutative geometry.

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2 Preliminaries on unbounded operators on Hilbert modules

Let B be a $\mathbb{Z}_2$ -graded C^* -algebra. Recall that a $\mathbb{Z}_2$ -graded Hilbert B -module E is a vector space equipped with a $\mathbb{Z}_2$ -graded right action $E \times B \rightarrow E$ and with a B -valued inner product $(\cdot|\cdot): E \times E \rightarrow B$, such that E is complete in the corresponding norm. The endomorphisms $\text{End}_B(E)$ are the adjointable linear operators $E \rightarrow E$, and the set $\text{End}_B^0(E)$ of compact endomorphisms is given by the closure of the finite rank operators. For a detailed introduction to Hilbert modules and $\mathbb{Z}_2$ -gradings, we refer to [Bla98, Lan95].

2.1 Non-symmetric operators

In this section, we describe our approach to dealing with non-symmetric operators, namely by studying the real and imaginary parts of such operators. Let us start with a useful lemma regarding the ‘combined graph norm’ of two closed operators on the intersection of their domains.

Lemma 2.1. *Let E be a right Hilbert B -module with inner product $(\cdot|\cdot)$. Let S and T be closed regular operators on E such that $\text{Dom } S \cap \text{Dom } T$ is dense in E . Then $\text{Dom } S \cap \text{Dom } T$ is a right Hilbert B -module with the inner product*

$$(\phi|\psi)_{S,T} := (\phi|\psi) + (S\phi|S\psi) + (T\phi|T\psi),$$

and the corresponding norm $\|\psi\|_{S,T}^2 = \|(\psi|\psi)_{S,T}\|_B$.

Proof. We need to show that $\text{Dom } S \cap \text{Dom } T$ is complete in the norm $\|\cdot\|_{S,T}$. Since S is closed, we know that $\text{Dom } S$ is complete for the graph norm $\|\cdot\|_S$ corresponding to the inner product

$$(\phi|\psi)_S := (\phi|\psi) + (S\phi|S\psi),$$

and a similar statement holds for $\text{Dom } T$. The inequalities

$$\frac{1}{2}(\psi|\psi)_S + \frac{1}{2}(\psi|\psi)_T \leq (\psi|\psi)_{S,T} \leq (\psi|\psi)_S + (\psi|\psi)_T$$

show that convergence in the norm $\|\cdot\|_{S,T}$ is equivalent to convergence in both graph norms $\|\cdot\|_S$ and $\|\cdot\|_T$. Denote by $\mathcal{W}_S$ (respectively $\mathcal{W}_T$) the closure of $\text{Dom } S \cap \text{Dom } T$ in the norm $\|\cdot\|_S$ (respectively $\|\cdot\|_T$). Then the closure of $\text{Dom } S \cap \text{Dom } T$ in the norm $\|\cdot\|_{S,T}$ is contained in the intersection of $\mathcal{W}_S$ and $\mathcal{W}_T$. Since $\mathcal{W}_S \subset \text{Dom } S$ and $\mathcal{W}_T \subset \text{Dom } T$, this intersection $\mathcal{W}_S \cap \mathcal{W}_T$ is contained in, and hence equal to, $\text{Dom } S \cap \text{Dom } T$, so we conclude that $\text{Dom } S \cap \text{Dom } T$ is complete in the norm $\|\cdot\|_{S,T}$. $\square$

In what follows, we will consider a closed regular operator $\mathcal{D}$ on a right Hilbert B -module E , such that $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is dense in E . The above lemma then tells us that $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is a Hilbert B -module with the inner product $(\cdot|\cdot)_{\mathcal{D},\mathcal{D}^*}$.

Definition 2.2. Let $\mathcal{D}$ be a closed regular operator on a Hilbert B -module E , such that $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is dense. We define the *real and imaginary parts* of $\mathcal{D}$ by setting

$$\text{Re } \mathcal{D} := \frac{1}{2}(\mathcal{D} + \mathcal{D}^*), \quad \text{Im } \mathcal{D} := -\frac{i}{2}(\mathcal{D} - \mathcal{D}^*),$$

on the initial domain $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$. Since these operators are densely defined and symmetric, they are closable, and we denote their closures by $\text{Re } \mathcal{D}$ and $\text{Im } \mathcal{D}$ as well. Furthermore, we define the ‘Wick rotations’ of $\mathcal{D}$ by

$$\mathcal{D}_+ := \text{Re } \mathcal{D} + \text{Im } \mathcal{D}, \quad \mathcal{D}_- := \text{Re } \mathcal{D} - \text{Im } \mathcal{D},$$

on the initial domain $\text{Dom } \text{Re } \mathcal{D} \cap \text{Dom } \text{Im } \mathcal{D}$. The term ‘Wick rotation’ is borrowed from physics, and its use is motivated by Proposition 4.8.

Lemma 2.3. *Let $\mathcal{D}$ be a closed regular operator on a right Hilbert B -module E , such that $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is dense in E . Then the norms $\|\cdot\|_{\mathcal{D}, \mathcal{D}^*}$, $\|\cdot\|_{\text{Re } \mathcal{D}, \text{Im } \mathcal{D}}$, and $\|\cdot\|_{\mathcal{D}_+, \mathcal{D}_-}$ (defined as in Lemma 2.1) are equivalent on $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$.*

Proof. An elementary calculation shows that we have the equalities

$$(\phi|\psi)_{\text{Re } \mathcal{D}, \text{Im } \mathcal{D}} = \frac{1}{2}(\phi|\psi) + \frac{1}{2}(\phi|\psi)_{\mathcal{D}, \mathcal{D}^*}, \quad (\phi|\psi)_{\mathcal{D}_+, \mathcal{D}_-} = (\phi|\psi)_{\mathcal{D}, \mathcal{D}^*},$$

from which it follows that the three norms $\|\cdot\|_{\mathcal{D}, \mathcal{D}^*}$, $\|\cdot\|_{\text{Re } \mathcal{D}, \text{Im } \mathcal{D}}$, and $\|\cdot\|_{\mathcal{D}_+, \mathcal{D}_-}$ are equivalent. $\square$

Lemma 2.4. *Let $\mathcal{D}$ be a closed regular operator on a Hilbert B -module E , such that $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is dense. If $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is a core for both $\mathcal{D}$ and $\mathcal{D}^*$, then $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^* = \text{Dom } \text{Re } \mathcal{D} \cap \text{Dom } \text{Im } \mathcal{D}$, $\mathcal{D} = \text{Re } \mathcal{D} + i \text{Im } \mathcal{D}$, and $\mathcal{D}^* = \text{Re } \mathcal{D} - i \text{Im } \mathcal{D}$.*

Proof. The operators $\text{Re } \mathcal{D}$ and $\text{Im } \mathcal{D}$ are initially defined on $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$, so the inclusion $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^* \subset \text{Dom } \text{Re } \mathcal{D} \cap \text{Dom } \text{Im } \mathcal{D}$ is obvious. Suppose that $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is a core for both $\mathcal{D}$ and $\mathcal{D}^*$. Consider the closed operator $\tilde{\mathcal{D}}$ defined as the closure of $\text{Re } \mathcal{D} + i \text{Im } \mathcal{D}$ on the initial domain $\text{Dom } \text{Re } \mathcal{D} \cap \text{Dom } \text{Im } \mathcal{D}$. Obviously, $\tilde{\mathcal{D}}$ and $\mathcal{D}$ agree on $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$, and since this domain is a core for $\mathcal{D}$, it follows that $\tilde{\mathcal{D}}$ is an extension of $\mathcal{D}$, and we have $\tilde{\mathcal{D}}^* \subset \mathcal{D}^*$. However, on $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ both $\tilde{\mathcal{D}}^*$ and $\mathcal{D}^*$ are given by $\text{Re } \mathcal{D} - i \text{Im } \mathcal{D}$, and since this domain is a core for $\mathcal{D}^*$, it follows that $\mathcal{D}^* \subset \tilde{\mathcal{D}}^*$. Hence $\mathcal{D}^* = \tilde{\mathcal{D}}^*$ and therefore $\mathcal{D} = \tilde{\mathcal{D}}$. By construction of $\tilde{\mathcal{D}}$ we have the domain inclusions

$$\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^* \subset \text{Dom } \text{Re } \mathcal{D} \cap \text{Dom } \text{Im } \mathcal{D} \subset \text{Dom } \tilde{\mathcal{D}} \cap \text{Dom } \tilde{\mathcal{D}}^*.$$

Since we have shown that $\mathcal{D} = \tilde{\mathcal{D}}$, we conclude that these inclusions are equalities. $\square$

Definition 2.5. Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be closed, regular and symmetric operators on a Hilbert B -module E , such that $\text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$ is dense in E . We define the *reverse Wick rotation* of the pair $(\mathcal{D}_1, \mathcal{D}_2)$ as the closure of

$$\mathcal{D} := \frac{1}{2}(\mathcal{D}_1 + \mathcal{D}_2) + \frac{i}{2}(\mathcal{D}_1 - \mathcal{D}_2)$$

on the initial domain $\text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$ (note that $\mathcal{D}$ is closable, because it is the sum of a symmetric and an anti-symmetric operator, which ensures that $\mathcal{D}^*$ is densely defined).

Remark 2.6. We emphasise that the reverse Wick rotation $\mathcal{D}'$ of the pair $(\mathcal{D}_2, \mathcal{D}_1)$ is *not* equal to the reverse Wick rotation of $(\mathcal{D}_1, \mathcal{D}_2)$, but they are related to each other: $\mathcal{D}'$ is the closure of the restriction of $\mathcal{D}^*$ to $\text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$. In other words, $\mathcal{D}^*$ is a closed extension of the closure of $\mathcal{D}'$, and they are equal if and only if $\text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$ is a core for $\mathcal{D}^*$.

Lemma 2.7. *Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be closed, regular and symmetric operators on a Hilbert B -module E such that $\text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$ is dense in E . Let $\mathcal{D}$ be the reverse Wick rotation of $(\mathcal{D}_1, \mathcal{D}_2)$. Then the norms $\|\cdot\|_{\mathcal{D}, \mathcal{D}^*}$, $\|\cdot\|_{\text{Re } \mathcal{D}, \text{Im } \mathcal{D}}$, and $\|\cdot\|_{\mathcal{D}_1, \mathcal{D}_2}$ are all equivalent on $\text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$.*

Proof. Let us write $\mathcal{E} := \text{Dom } \mathcal{D}_1 \cap \text{Dom } \mathcal{D}_2$. The operators $\mathcal{D}_1 \pm \mathcal{D}_2$ are symmetric on $\mathcal{E}$, so the domain of $\mathcal{D}^*$ also contains $\mathcal{E}$ (and in particular $\text{Dom } \mathcal{D} \cap \text{Dom } \mathcal{D}^*$ is dense). For $\psi \in \mathcal{E}$ we can then write

$$\mathcal{D}^* \psi = \frac{1}{2}(\mathcal{D}_1 + \mathcal{D}_2)\psi - \frac{i}{2}(\mathcal{D}_1 - \mathcal{D}_2)\psi.$$

Hence on the initial domain $\mathcal{E}$ we can write $\mathcal{D}_1 = \mathcal{D}_+$, $\mathcal{D}_2 = \mathcal{D}_-$, $\text{Re } \mathcal{D} = \frac{1}{2}(\mathcal{D}_1 + \mathcal{D}_2)$, and $\text{Im } \mathcal{D} = \frac{1}{2}(\mathcal{D}_1 - \mathcal{D}_2)$. From Lemma 2.3 it then follows that the norms $\|\cdot\|_{\mathcal{D}, \mathcal{D}^*}$, $\|\cdot\|_{\text{Re } \mathcal{D}, \text{Im } \mathcal{D}}$, and $\|\cdot\|_{\mathcal{D}_1, \mathcal{D}_2}$ are equivalent on $\mathcal{E}$. $\square$

2.2 Almost (anti-)commuting operators

Almost (anti-)commuting operators were considered by Mesland in [Mes14], and later generalised by Kaad and Lesch in [KL12], for the construction of the unbounded Kasparov product. Almost anti-commuting operators play an important role later in proving that the Wick rotations of indefinite Kasparov modules are (genuine) Kasparov modules. In this section, we recall the results from [KL12], and prove a few further consequences.

Definition 2.8 (see [KL12, Assumption 7.1]). Let S and T be regular self-adjoint operators on a Hilbert A -module E such that

- 1) there exists a submodule $\mathcal{E} \subset \text{Dom } T$ which is a core for T , and
- 2) for each $\xi \in \mathcal{E}$ and for all $\mu \in \mathbb{R} \setminus \{0\}$ we have the inclusions

$$(S - i\mu)^{-1}\xi \in \text{Dom } S \cap \text{Dom } T \quad \text{and} \quad T(S - i\mu)^{-1}\xi \in \text{Dom } S.$$

The pair (S, T) is called an *almost commuting pair* if in addition

- 3) The map $[S, T](S - i\mu)^{-1}: \mathcal{E} \rightarrow E$ extends to a bounded operator in $\text{End}_A(E)$ for all $\mu \in \mathbb{R} \setminus \{0\}$.

Similarly, the pair (S, T) is called an *almost anti-commuting pair* if instead of 3) we have

- 3') The map $\{S, T\}(S - i\mu)^{-1}: \mathcal{E} \rightarrow E$ extends to a bounded operator in $\text{End}_A(E)$ for all $\mu \in \mathbb{R} \setminus \{0\}$.

These conditions are often summarised by simply saying that $[S, T](S - i\mu)^{-1}$ (or $\{S, T\}(S - i\mu)^{-1}$) is well-defined and bounded.

Lemma 2.9. *Let (S, T) be a pair of regular self-adjoint operators on a Hilbert module E satisfying 1) and 2) of Definition 2.8. Then S is essentially self-adjoint on $\text{Dom } S \cap \text{Dom } T$.*

Proof. By assumption we have $(S \pm i)^{-1}(\xi) \in \text{Dom } S \cap \text{Dom } T$ for all $\xi \in \mathcal{E}$, where $\mathcal{E}$ is dense in E . Since S is self-adjoint, the operator $(S \pm i)^{-1}$ is bounded and has range $\text{Dom } S$, which is dense in E . Hence $(S \pm i)^{-1}\mathcal{E}$ is also dense in E , from which it follows that $\text{Dom } S \cap \text{Dom } T$ is dense in E , so the operator $S|_{\text{Dom } S \cap \text{Dom } T}$ is symmetric and densely defined on $\text{Dom } S \cap \text{Dom } T$. Furthermore, the image of $(S \pm i)|_{\text{Dom } S \cap \text{Dom } T}$ contains $\mathcal{E}$ and is therefore also dense, which implies that $S|_{\text{Dom } S \cap \text{Dom } T}$ is essentially self-adjoint. $\square$

Given two regular self-adjoint operators S and T on a Hilbert A -module E , we consider two new operators on $E \oplus E$ given by

$$\tilde{S} := \begin{pmatrix} 0 & iS \\ -iS & 0 \end{pmatrix}, \quad \tilde{T} := \begin{pmatrix} 0 & T \\ T & 0 \end{pmatrix},$$

with domains $\text{Dom } \tilde{S} = (\text{Dom } S)^{\oplus 2}$ and $\text{Dom } \tilde{T} = (\text{Dom } T)^{\oplus 2}$. One easily calculates that

$$[\tilde{S}, \tilde{T}] = i \begin{pmatrix} [S, T] & 0 \\ 0 & -[S, T] \end{pmatrix}, \quad \{\tilde{S}, \tilde{T}\} = i \begin{pmatrix} \{S, T\} & 0 \\ 0 & -\{S, T\} \end{pmatrix},$$

whenever these operators are defined. Hence this ‘doubling trick’ allows us to easily switch between almost commuting and anti-commuting operators.